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the archeological heritage of easter island

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by William MULLOY

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THE ARCHAEOLOGICAL HERITAGE OF

RAPA NUI (EASTER ISLAND)

A report prepared for The United Nations Educational, Scientific, and Cultural Organization

Being a summary of the nature and significance of the archaeological monuments of the island and proposed means and methods for their conservation and restoration in order to record permanently as an island-wide museum an important segment of Polynesian prehistory

William Mulloy
Professor of Anthropology
University of Wyoming
United States of America

and

Gonzalo Figuerao
Archaeologist
University of Chile
Republic of Chile

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PREFACE

The following exposition was prepared at the request of The United Nations Educational, Scientific and Cultural Organization in fulfillment of Consultant Contract No. CLT - 0768 and as a part of a wider investigation leading to a general development plan for the island of Rapa Nui.

The contract was carried out by William Mulloy, Professor of Anthropology at the University of Wyoming. Mulloy enlisted the cooperation of Gonzalo Figueroa, archaeologist from the University of Chile, who because of his extensive experience in archaeological investigation on the island, possessed unique knowledge of the problems and archaeological monuments involved.

Figueroa and Mulloy visited the island during a fifteen-day period between March 25 and April 7, 1966. During this time much information was obtained for

inclusion in this report, though, in the case of both writers, previous experience of approximately two years duration in archaeological investigation and restoration on the island was drawn upon heavily.

Charles E. Peterson, Architectural Historian, Restorationist and Planner with long experience with the United States National Park Service and in Pacific island development, and Raul Bulnes, Architect from the Ministry of Public Works of Chile accompanied the writers to the island and the investigations were carried out with their cooperation. Peterson provided many excellent suggestions bearing on the approach of this report and its content and Bulnes prepared the map which is here included. Peterson has prepared a report presenting preliminary plans and analysing the broader aspects of the island development prospects of which the present contribution may be considered to be an amplification of a single aspect.

The time on the island was devoted to daily trips to various localities by jeep and on horseback while

returning to the town of Hangaroa each evening. All of the most important localities were visited and an adequate sample of the archaeological monuments was examined.

The writers were deeply indebted to Mr. Enrique Rogers, recently appointed Governor of Isla de Pascua (Rapa Nui) and to Mr. Francisco Oyarzun of the Ministry of Foreign Affairs of Chile for enthusiastic encouragement as well as a wealth of background information, maps and many other kinds of indispensible aid and cooperation without which it would have been impossible to write this report. Commander Arne Arentsen, present Military Governor of the island went far out of his way to make our stay pleasant and productive by providing excellent housing and transportation as well as many other vital services. Father Sebastian Englert, the island priest, made available without reservation his unique knowledge of the local archaeology and ethnology and of local historical developments. One and all of the people of the island were enthusiastic and interested in our work

and gave freely of services and encouragement far beyond what might have been expected. The University of Wyoming provided a part of Mulloy's services and The Fulbright Commission his transportation to and from continental Chile. The Navy of Chile provided transportation to and from the island for the whole party. Dr. Carlyle Smith of the Department of Anthropology of The University of Kansas, and Mr. Edwin Ferdon of The Arizona State Museum, University of Arizona, read the manuscript and offered suggestions. They are not responsible for any of the errors or shortcomings of the report. The writers are deeply indebted to all of these individuals and institutions and to many others, whose names cannot be mentioned here, for the aid they have offered. The views expressed are those of the authors and not necessarily those of Unesco.

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Archaeological Map of Rapa Nui in Attached Pocket

THE ARCHAEOLOGICAL HERITAGE OF RAPA NUI (EASTER ISLAND)

PART ONE - BACKGROUND

Description of the Island

Rapa Nui (Easter Island) is the easternmost island of Polynesia and is located at approximately 27°S Lat. and 109°W Long., being about 2300 miles west of the coast of Chile. It is one of the inhabited localities in the world most isolated by the sea, the nearest populated islands being Pitcairn about 1300 miles to the west and the Juan Fernandez group about 1600 miles to the southeast.

The island is formed by the triangular summit of a mountainous uplift which rises some 1400 feet above the general floor level of the Pacific. (See map). Though some coral grows irregularly near the shore, there is no barrier reef and in most localities shores slope sharply to considerable sea depths. The exposed land is composed entirely of volcanic materials and is a

cluster of volcanic cones. It owes its triangular shape to three major vents of which one lies at each apex. The intersecting and conforming slopes of these cones form the central part of the island which is composed of beds of scoria and tuff and extensive lava flows together with many areas of excellent soil resulting from the decomposition of these materials. The central part thus has the aspect of relatively low rolling country through which are distributed many parasitic cones sometimes of considerable size.

The southwestern cone, Rano Kao, has a large, deep crater with an extensive fresh water lake almost entirely covered with a thick floating mat of totora reeds. Maunga Terevaka, the northwestern and highest cone, has an insignificant dry crater in its summit while in its southeastern slope is a larger crater containing a small fresh water lake also covered by a mat of floating totora. This is called Rano Arci. The eastern apex is formed by a volcano called Poike which also has in its summit an insignificant dry crater. Near the south shore

and the foot of Poike lies the larger crater of Rano Raraku which is a prominent terrain feature. It also contains an extensive fresh water lake with a floating mat of totora. (See Photos, Nos. 17-18).

Large parts of the shore line have been deeply eroded by the sea to form high cliffs especially at the apexes. Small beaches are found only at Hangaroa, Anakena, Ovahe, and Hanga Hoonu. The coasts are thus principally steep and rocky with few large indentations and few good anchorages. Along its longest side, Rapa Nui is about twenty-three kilometers long. Its widest breadth is about twelve kilometers. In all, there are about 117 square kilometers of area or about 12,000 hectares.

Climate is warm and moderately humid. Average temperatures are around 21°C. with morning and evening temperatures falling to around 18°C. and midday temperatures rising to 23°C. Average minimum temperature is around 16°C. The warmest month is February and the coldest is July but the difference is only about 5°C.

Average rainfall is about 125 centimeters per year. In some years, it has risen to over 178

centimeters and occasionally has fallen as low as 71 centimeters. Rain tends to fall in brief local showers which are intense and cover only a part of the island. Occasionally there are longer rains but these are not common. Although more rain falls in winter than in summer there is no clear cut seasonal rhythm. The average cloud cover is about 5/8 with the May to October period having slightly more cloud cover than the rest of the year. Wind of moderate velocity blows more or less steadily from the east during most of the year. From June to September, the winds tend to become variable and to blow from between northeast and northwest.

Rapa Nui lacks permanent flowing streams and, depending upon the nature of its former vegetation, may never have had them. If earlier conditions were similar to those of today, the only fresh water that would have been available to prehistoric inhabitants would have come from the three previously mentioned crater lakes where an ample supply is present today and a large number of smaller sources that were probably of basic importance in the old

way of life. Among these minor but numerous sources are subterranean pools to be found in hollow volcanic tubes in several parts of the island. Some of these at least appear to maintain during most years enough domestic water for fairly large communities. An example of one of these is Vai Teka inland of the central part of the west coast. (See map). Also in various localities are what appear to be catch basins or crevices in the lava into which water flows. Some of these would provide amply for the needs of small communities. Examples are Vai Tara Kai Ua inland west of Anakena and Ahu Hekii on the eastern part of the north coast. (See map). Also along the shores at many places fresh water seeps out at sea level and can be obtained at low tide. In many parts of the island may be seen taheta or artificially carved and sometimes covered basins which catch rain water. These were probably made for this purpose and collect and hold a considerable amount of water for a surprising time after rains. Many natural basins in stone surfaces perform the same function. In some localities relatively large basins in earth

surfaces hold a considerable quantity of water after rains. Some are surrounded by stone pavements. An example is Vai Arepa northeast of Hangaroa. With these sources, a fairly large and scattered population accustomed to use fresh water only for essential purposes could have been maintained without flowing streams.

The nature of the original natural plant cover of Rapa Nui is a question still being studied and to which no definitive answer can be given at present. Pollen samples recovered from the crater lakes have been varied and suggest the possibility of a much more luxurious plant cover than that known in historic times. The results of this work have not yet been published but they have important significance for the future of the island - if a more luxurious vegetation once grew it might be made to grow again. This view is strongly supported by the success which has attended the planting and cultivation of a variety of trees and plants.

Skottsberg (1920, p. 494 and following) suggests that the original plant cover was a park-like savanna

with groves of toromiro trees (sophora toromiro) and shrubs in sheltered places interspersed with grass land in more exposed localities. He explained the paucity of vegetation under conditions suitable for more luxurious growth by failure of seeds to survive the long sea journey. The only tree that arrived according to this view was the Sophora toromiro. Along with it came the shrub Lycium carolinian, while the rest of the vegetation was predominately herbacious.

If, as the archaeological evidence seems to indicate, a small group of colonists arrived before 800 A.D. (perhaps considerably before), they might well, as they increased in numbers, have cut down more and more of the natural vegetation for plantations and for other purposes until the immediately prehistoric period of over-population which is described in local traditions produced the barren landscape which greeted the earliest explorers. These at least were unanimous in their description of extensive garden cultivation widely seen with only patches of brushwood in the distance.

Much of the formerly cultivated land appears to have reverted to grass land after the sudden decrease in population resulting from the slave raids of 1859-63. Several kinds of natural grasses appear to have grown and others were probably introduced. The commonest species today and the characteristic feature of most of the landscape is a tough short grass locally called heri hoi (Sporobolus indicus).

Another noteworthy plant is the previously mentioned totora (Scirpus riparius) which grows thickly as floating masses in crater lakes and was useful for making mats and similar textiles.

The prehistoric population introduced a good many plants among the more important of which were several kinds of sweet potatoes, taro, bananas, yams, gourds, sugar cane, paper mulberry, and hau hau (Hibiscus tiliaceous). Coconut may have been introduced and later died out. Breadfruit is not known to have been present.

A wider variety of trees and smaller plants have been introduced in historic times and most of

them do well. Some of these amplified considerably the variety of food available and some parts of the island at least have begun to assume a verdant appearance quite different from the island's former characteristic barrenness. Sharp improvements in this respect were evident between the 1955 and 1966 visits of the writers.

Animals include numerous species of typical Pacific marine forms among which many varieties of fish form an important source of food. Many species of long range sea birds are also present and they and their eggs have also provided a food source. Of mammalian species, there are only those introduced by man. The chicken, and the kio'e or Polynesian rat, both used for food, were brought in prehistoric times and still survive. The dog and pig, other Polynesian animals that might have been expected, are not known to have been introduced before historic times though they are now present. Two kinds of small lizards called moko also arrived in prehistoric times possibly as stowaways on some aboriginal voyage.

Cats which have since gone wild, rabbits and goats which have since died out, dogs, horses, sheep, pigs, cattle and several varieties of birds are the most important species that have been introduced in historic times. (Part of the information in the foregoing section was obtained from Wright and Diaz, 1962, which contains a useful and detailed general description of the island).

Culture History of the Island

The earliest radio-carbon date of unquestioned reliability is that from Ahu No. 2 at Vinapu and is 857 A.D. ± 200. An earlier date from the Poike Ditch of 380 A.D. ± 100 has been seriously questioned and opinion must be reserved on it. It thus appears probable that during the 9th century A.D., a human population was already on the island. That they may have been resident there a long while before this is suggested by the fact that the date is associated with a large open air altar called ahu that seems unlikely to have been among the earliest structures built.

While it is impossible to deal in this short

account with the extremely controversial and complicated question of the origins of this population and whether these people were Polynesians or American Indians or some sort of mixture of both, the significant fact seems to be that a probably small group, possibly reenforced by other arrivals from time to time, did appear on the island before the date indicated and gradually increased in numbers until they included several thousand people at the time of their first historic contact in 1722. At the time of historic contact they spoke, and still speak, a Polynesian language.

Whatever their origin may have been, and this question will not be resolved fully without much additional archaeological investigation on the island and elsewhere, the most remarkable thing about them is what they were able to achieve in a locality of almost complete cultural isolation and one with relatively limited resources. Their cultural record is among the more remarkable examples of human capacity and ingenuity in isolation.

While it is reasonable to assume that the

earliest immigrants to Rapa Nui were first exclusively concerned, and probably for considerable time, with such life basics as development of food and shelter sources and the establishment of a working economy, we have no good archaeological evidence as yet of this early development period. It does seem likely, however, that some time before 800 A.D. these people began to develop by far the most remarkable religious building compulsion known in the whole Pacific. Perhaps it was encouraged by an economy that became so effective that considerable amounts of leisure resulted. The ease with which food crops can be grown today suggests this.

The distribution of the examples of this architecture suggest that at this time the people were scattered widely along the coasts and adjacent interior areas of the island in relatively small independent or partially independent kin groups of some sort. Local traditions which have survived, suggest this sort of arrangement. Each of these groups possessed its own religious

structure or structures which were variants of a widely diffused kind of Polynesian marae locally called ahu. (See Photograph No. 2). Though there is much variation among these and many have undergone complicated sequences of rebuilding so that their earlier forms are obscured, presently available evidence suggests that the most frequent type consisted of a long, narrow, dry masonry platform with a sloping ramp on one side which might or might not have a slightly elevated central platform and lateral wings. Facing the ramp was frequently an artificially flattened plaza beyond which was an irregular row of long narrow boat shaped houses of thatch with precisely fitted stone foundations, which are traditionally said to have been occupied by members of a priesthood. The remains of most of these ahu are located along the shores of the island with plazas on the inland side, while some ahu occupy inland locations. Typically inland of the shoreside ahu were the domestic establishments of other members of the kin group which sometimes extended a considerable distance into the

interior. Thus the typical settlement pattern of a group appears to have included a segment of coast and its adjacent inland zone.

We do not know whether or not the earliest of the ahu had statues on their platforms, but it is clear that at some point in their history, statues began to be placed on them in rows of from one to fifteen facing toward the plazas and that in general, through time, they increased in size and stylization until a highly uniform type developed. (See Photograph No. 2). Traditionally, these statues are said to have represented ancestors and to have possessed supernatural power.

Within the framework of this general plan, the ahu ruins vary greatly in size and individual characteristics and typically complicated sequences of rebuilding are present in which characteristics have been changed and innovations added. Some are remarkable for their size (platforms over 150 M. long), others for the number or size of statues (one about ten meters high), others for their height and the size of the stones in their platform walls, and still

others for the precision of the stone fitting in their masonry. Many of these individualized characteristics undoubtedly have chronological significance. Others may have resulted from individual preferences of kin groups and a sense of competition among them. It is likely that the size and ostentation of features of many ahu reflected the numbers and capacity of the kin groups that built them. Rebuilding in many cases may have resulted from increased capacity of a kin group and a desire for a more impressive structure of structures. (See Photographs Nos. 7-16).

Though some statues were carved from stone obtained in a variety of locations, it appears that early in the statue building period, a single source became highly dominant. This was the southwest exterior and interior slopes of the volcanic cone of Rano Raraku. Here intensive quarry operations were carried out with statues being almost completely carved while still attached to the living tuff. When detached, they were slid down the slopes of the volcano on their backs and transported to the

ahu by means that can be speculated upon but are not known. Traditions state that the statues walked to their ahu locations.

Many statues remain partly finished in the quarry and examples may be seen in every stage of construction. Many statues stand at the base of the exterior talus of the quarry and some fewer in the interior of the crater itself. Whether these were awaiting transport to ahu or were intended to remain standing in their present locations is not clear at present. (See Photographs Nos. 17-26).

It seems likely that this architectural and artistic activity reached its greatest intensity at a time shortly before historic contact when the culture as a whole reached its optimum population and greatest economic productivity. These factors contributed to a maximum organization and achievement of the religious building compulsion with production of the largest and most impressive statues and ahu.

Such an effective equilibrium rarely lasts

long and it seems likely that this one began to be destroyed shortly before the beginning of historic times by over-population resulting in a scarcity of food and eventually fomenting internal warfare, one of the most characteristic goals of which was the overthrowing of the statues of enemy groups and the destruction of their ahu. (See Photographs Nos. 10, 13-14). This warfare appears to have lasted well into historic times. It was attended by a serious cultural decline and a new use for the destroyed ahu. While present evidence suggests the likelihood that the dead were previously cremated in special structures associated with the ahu, (See Photograph No. 31), at this time tombs began to be built in the ramps and platforms of the destroyed ahu. Other cadavers were simply laid out on the ramps and among the fallen statues and covered with stones. In some cases, destroyed ahu were almost entirely covered with stones in this manner so that few vestiges of their formerly impressive masonry and architectural characteristics remain visible. Many ahu began to

assume the appearance of long narrow piles of stones with sloping sides, highest at the center and with pointed ends. At this time, new piled stone ahu of similar appearance were built in new locations to provide additional burial places.

The first European explorers thus saw a culture with a numerous population suffering from internal war and which, to a considerable extent, was living among the ruins of its former spectacular architectural and artistic accomplishments.

The first European to visit the island was the Dutchman Jacob Roggeveen, who arrived on Easter Sunday, 1722. In 1770, the Spaniard Felipe Gonzalez took possession of the island for Spain. In 1774, the Englishman James Cook and in 1786, the Frenchman J-F. de G. La Perouse paid brief visits. From this time, gradually increasing numbers of visitors have brought the islanders into progressively more intense contacts with the outside world.

The next cultural event of devastating importance to plague the islanders was a series of slave raids which began in 1805 when the schooner

Nancy, out of New London (U.S.A.), carried away twenty-two people as slaves. The most important raids were by Peruvians between 1859 and 1862 when about a thousand islanders were carried off to work on the guano islands off the coast of Peru. Bishop Jaussen of Tahiti protested this crime to the French Minister at Lima and the survivors, numbering about a hundred, were returned. All but fifteen died of smallpox on the return voyage and on arrival, these spread the disease on the island. The population was quickly reduced to a few hundred. From this small remnant, the present islanders are descended.

The dead appear to have included all of the maori rongo rongo or the local priesthood and as a result much important religious and other esoteric information, including the technique of reading the Rapa Nui written language, was lost. A considerable number of the wooden tablets bearing this script survive, but they remain undeciphered to this day.

Christianity was established on the island in 1864 by Lay Brother Eugéne Eyraud of the Congregation des Sacres Coeurs of Picpus. It has continued in a

complicated history to the present time.

In 1870, the first commercial exploitation of the island was begun by Jean Baptiste Dutroux-Bornier who settled on the island as a sheep rancher. Sheep and other cattle have continued to be the important industry on the island down to the present time. In 1888, the island was annexed by Chile through the efforts of Captain Policarpo Toro Hurtado. The commercial resources of the island, being principally sheep and to a lesser extent other cattle, were leased to various private companies until 1954 when their administration was taken over by the Chilean Navy. In June of 1966, it is proposed to replace the naval administration of the island with a civil one.

Since the smallpox epidemic resulting from the slave raids, the population has gradually increased until the islanders now number more than 1200 of which between eight and nine hundred are on the island. At present, all live in the village of Hangaroa, on the west coast.

Previous Archaeological Work

Although a great many people have written of the superficial appearance of the archaeological monuments of the island with varying degrees of objectivity, there have been few serious archaeological investigations. In 1889, W. J. Thomson spent eleven days on the island and conducted a survey of archaeological remains (Thomson, 1889). In 1914-15, an expedition was led by Mrs. Katherine Routledge who, with her party, remained on the island for seventeen months. Though the major records of this expedition were lost, the popular book that was published contains a great deal of useful and accurate information. Her subsurface archaeology was confined largely to excavations around the bases of statues in the quarry at Rano Raraku (Routledge, 1919). In 1934-35, an expedition led by Henri Lavachery and Alfred Métraux spent five months on the island. Lavachery's principal contribution was the recording of a vast number of petroglyphs and a number of surface observations of monuments, (Lavachery, 1932-54).

Métraux produced an excellent ethnology and many observations relating to archaeological remains, (Métraux, 1936-57).

In 1948 Father Sebastian Englert published a volume on the island's archaeology, ethnology, history and linguistics resulting from his thirteen years of investigation during residence as the island priest, (Englert, 1948). A second and much enlarged edition of this book including the results of subsequent studies up to the present is now in press.

The most comprehensive archaeological investigation was carried out by an expedition under the direction of Thor Heyerdahl which was on the island for a little over five months in 1955-56. Six archaeologists were present (Carlyle Smith, Eduardo Sanchez, Edwin Ferdon, Gonzalo Figueroa, William Mulloy, and Arne Skjölsvold) and samples of many kinds of monuments were investigated, (Heyerdahl and Ferdon, Eds., 1961).

In 1958, Thomas Bartel and Ruperto Vargas spent several months on the island. Bartel worked

principally with the local linguistics and the problem of translation of Rongo Rongo tablets, and also collected a good deal of traditional information, (Bartel, 1958-62). Vargas investigated several archaeological sites.

In 1960, Gonzalo Figueroa and William Mulloy spent about a year on the island. Their principal accomplishment was the complete investigation and restoration of Ahu Akivi. (See Photographs Nos. 1-6). The investigation was carried out to cast further light upon the architectual sequence established by the Heyerdahl expedition. The restoration was a pilot project to demonstrate that ahu restoration could be done at reasonable cost and that it should be an important part of a future program of conservation of monuments. (Figueroa and Mulloy, 1960, Mulloy and Figueroa, 1961, 63).

In 1965, William Mulloy spent four months on the island. No excavation was done. The major work was the measurement of the azimuths of the facades of about 300 ahu in order to determine how many of them were oriented to the direction of the rising

or setting sun at the equinox or solstices or had other significant stellar orientation.

Although a number of people have engaged in archaeological work on Rapa Nui, the vast bulk of the monuments remain uninvestigated, many of the most important problems remain unstudied or controversial, and a thorough understanding of the nature of developments must await many years of future research.

Scientific Values

Future investigation of the prehistory of Rapa Nui is of first order importance to the understanding of Polynesian prehistory as a whole. The remarkable wealth of monuments would be, even if they had only incidental significance, of unique importance. The physical manifestations of culture history are here in abundance to be studied. While the people of many other islands worked in highly perishable wood, here there was a strong tendency to work in durable stone. The curious religious building compulsion led to the production of fantastic numbers of objects of art and architectural

monuments. The stylized motifs found among these, and their changes through time, together with the physical stratigraphy which relates them to each other, provide a wealth of the basic counters with which historical interpretations are made. There is no other island in the whole of the Pacific that approaches Rapa Nui in this respect.

Moreover, Rapa Nui is the easternmost island in Polynesia. It represents one of the end points of the complicated paths followed by migrants across the Pacific. The characteristics and achievements of its inhabitants represent, in a sense, the ultimate diffusion of the Polynesian migration.

With its easternmost position, it is also adjacent to the New World and it thus represents a most significant locality in the solution of the controversial problem of New World contacts in eastern Polynesia. That some sort of contact has taken place from this source, is difficult to dismiss summarily. Its nature, time, and intensity involve questions that undoubtedly will find significant elements of their answers in Rapa Nui

prehistory.

Beyond these culture historical considerations, the island offers potential information which contributes to the general theoretical understanding of cultures in isolation. It is a well recognized principle of culture growth that civilization and cultural complexity are strongly encouraged by large populations and frequent outside contacts. Civilizations tend to be at crossroads and the simplest cultures at the ends of the earth. Yet, in profound violation of this principle in Rapa Nui, one of the more isolated areas of the world and with a population that never exceeded a few thousand, were found such ideas characteristic of complex cultures as a written language, (not known to be related to any other script); sophisticated interest in solar phenomena; highly stylized, gigantic, monolithic stone sculpture in hundreds of examples together with the engineering techniques necessary to move and erect it; and a well developed and varied religious architecture requiring organized, large group activity to build. The understanding of this

peculiarity has broad theoretical significance. At least a part of the answer may become apparent when Rapa Nui culture history is better understood.

Museum Values

Because of the remarkable number, size and variety of the archaeological monuments on Rapa Nui, there is no other island in the Pacific remotely approaching it in adaptability to be developed into an island-wide museum of Polynesian prehistory. This could be accomplished by a relatively modest long term project of archaeological conservation and restoration carefully organized to coordinate precisely with agricultural and other kinds of land use and with other projects for the development of other aspects of the resources of the island. Due to the fact that the greater number of monuments tends to be concentrated near the coasts and the best agricultural land tends to be in the interior, monument use of land would interfere little with other kinds of exploitation. Moreover, in addition to the obvious scientific value of such an island-wide museum, it would quickly become world famous

and, with transportation facilities available, would attract tourists in such numbers that land so used would quickly come to have much greater value than might be developed by any other means of exploitation.

As previously mentioned in the historical section, during a late prehistoric period of internal warfare, practically all of the most impressive archaeological monuments were deliberately destroyed and remain as disordered stone piles of little exhibition significance. The original stones are usually still at hand, however, and careful reassembly and re-erection of statues can reconvert them into impressive monuments. An example of this, is the previously mentioned Ahu Akivi which before investigation and restoration, appeared as a disordered group of fallen statues and fragments. After restoration, it became an impressive platform some 90 meters long and facing on a spacious plaza. (See Photographs Nos. 1 and 2). Hundreds of other ahu, many much larger and bearing gigantic statues, are capable of similar restoration.

While a more detailed description of monument types will be provided in later pages, it is worth

noting here that many other kinds of monuments can be similarly restored. There are over a thousand statues. There are remains of many scores of villages with house foundations of precisely cut basalt, while the ceremonial village of Orongo, high on the lip of the crater of Rano Kao, is especially important. There are uncounted platforms, ramps, stone chicken houses, clusters of circular stone enclosures, (presumably for growing crops), petroglyphs, masonry lined caves, stone towers, boundary markers, roads, walls, and a wide variety of unique monuments.

It would be expected that such a project would include an enclosed museum to house and exhibit the thousands of smaller objects that would result from the investigations, as well as suitable storage, library, laboratory and other facilities to make possible the carrying out of the field work.

PROPOSAL FOR A PROJECT OF ARCHAEOLOGICAL INVESTIGATION, CONSERVATION AND RESTORATION

PART TWO

Archaeological Inventory

As yet a remotely complete archaeological inventory of the island has never been attempted. The endless numbers of monuments and the time required to locate and describe them is so great that it is easy to understand why no investigator has ever dared to begin the task. The problem has been further complicated by the fact that, until 1965, no reasonably accurate map had been available on which the results of such an inventory could be recorded. A few attempts have been made to locate and enumerate the many ahu but because of incompleteness and lack of accurate maps, they are not adequate to the purposes of the present proposal. Thomson recorded 113 ahu, (1889, pp. 499-513), Routledge estimated 260 ahu, of which she appeared to have been able to identify between 160 and 190, (1919, pp. 230-231), Lavachery listed

183 ahu, (1935, pp. 284-287), and Englert counted 244 ahu, (1948, pp. 97-111, 515-533). Mulloy, in 1965, while using Englert's list and map as a guide for a study of ahu orientation recorded fifty-two additional ones. It is his opinion that there are probably at least fifty more beyond this on the island. Englert has numbered and recorded the locations of statues over the years and his list now includes over 800. As yet unrecorded examples and those that will appear in excavations will surely bring the eventual number well above 1000. No attempt has been made to locate or enumerate other classes of monuments except certain unique ones.

Clearly the production of a complete and precise inventory is one of the first tasks that must be completed. The Chilean Air Force published in 1965 advance proofs of a contour map at a scale of 1:10,000 and a complete coverage of aerial photographs at a scale of approximately 1:25000. These form the indispensable basis of an archaeological inventory. No small or easy task is represented and it must be expected to consume considerable

time. With relation to existing reference points on the island, the area should be divided into a large number of sectors - as an estimate about forty-five or fifty. Enlargements of the existing contour map to a scale of about 1:2,000 should be prepared. Each sector should be approached separately and all monuments in it recorded on the map. This can be accomplished only by close study of every square meter of the terrain and the recording of the monuments with a plane table or other instrument. Photographs and a summary description of each monument should be made and with assigned numbers a catalogue should be established. This should be done by a crew of professional surveyors and a photographer in co-operation with an archaeologist. The writers estimate that an archaeologist with a crew of professional assistants could finish this task in about a year and a half.

Such an inventory would form an important part of the basis for all future archaeological and conservation work. It would indicate precisely, in a way that cannot be remotely approached at present

which, often quite restricted and isolated, areas must be reserved for archaeological conservation purposes and which are available for road building, construction, agricultural and other purposes. As various kinds of land utilization are developing rapidly at present, it is essential that this inventory be available as soon as possible in order to avoid destruction of or serious damage to valuable monuments. As a number of roads and other kinds of construction have already been completed and many others are projected for the near future, as new land is continually being assigned to individuals for cultivation, and as private land titles will soon be established, it is clear that the inventory will be badly needed long before it could possibly be available.

It must be emphasized strongly that the map with this report does not represent an archaeological inventory. It is probably the best archaeological map of Rapa Nui that now exists, but it was made by the writers by transferring the locations of monuments from Englert's map, studying air photographs, and

making use of personal knowledge of monument locations. It most emphatically was not produced by a survey. The lines that define the archaeological zone are most provisional and arbitrary and by no means all-inclusive. In default of an inventory, these have been drawn to indicate in the most general terms only the zone where archaeological monuments are most common. Once an inventory is available, certain areas within this zone can be certified free of monuments and devoted to other purposes. On the other hand, many limited areas outside the zone will be found to have monuments of interest and will require small enclave reservations. It is impossible to be more precise with regard to this problem until the inventory is available. On the map are marked the very approximate locations of only most of the more important monuments - principally ahu. Most but not all of these are near the coast. The typical settlement pattern on this island includes one or more ahu near the coast. To landward of each ahu is a large plaza. To landward of the plazas are series of boat shaped houses, traditionally said to have

been occupied by a priesthood. Farther toward the interior, are found isolated domestic establishments in gradually diminishing numbers. These include house sites, habitation caves, chicken houses, agricultural enclosures and other features. In some cases, this pattern extends into the interior a considerable distance beyond the borders of the marked archaeological zone.

One, perhaps somewhat peripheral, investigation of great culture historical value should be carried out as part of the inventory survey. This is the recording of place names. Like many Polynesian islands, Rapa Nui is unusually rich in place names. Last year a Chilean naval officer visited the island to collect place names for the new map and recorded a large number that have contemporary map significance. These represent only a small part of the still known names. Many of the names have traditions associated with them that are known to a few old people who will die soon. It would be highly desirable to assign an individual, to be accompanied by several old islanders, to the survey party for the specific purpose of

recording these place names and the traditions associated with them in great detail. A publication of unique historical value would result.

Archaeological Investigation, Conservation, and Restoration

That the archaeological monuments of the island are of high economic value for purposes of tourism in addition to their scientific and historical value cannot be doubted. Their conservation thus merits the most serious and systematic attention. As a result of effects of prehistoric warfare, other vandalism, and time itself most are in bad condition and in serious need of well considered attention. Projected tourism makes some sort of conservation measures even more urgent, not only for the protection of the monuments themselves, but to expose and make them more understandable to visitors. Each class of monuments presents its own special problems of conservation and restoration and will be considered individually.

It is recommended most emphatically that all monuments should not be restored, at least at this

time. Some ahu especially and other monuments as well should be investigated, stabilized, and left substantially in their present condition to illustrate the effects of prehistoric demolition. Other examples should be superficially stabilized and not investigated. If the recent past is any index, archaeological techniques will be improved greatly and it will be possible for archaeologists of the future to learn much more from them than present methods permit. Though examples of all types should be so reserved, which monuments would be selected could be determined only by future detailed studies.

Ahu: In all there are probably about 350 ahu on the island. (See Photographs Nos. 1-2, 7-16). Probably about 150 have from one to fifteen now fallen statues. Platforms and ramps of these image ahu are of dry stone masonry with rubble fill. Masonry varies from large, precisely cut and fitted stones through large, carefully selected and fitted stones, to rather ill selected stone block construction. Although walls are frequently badly disassembled almost always the displaced stones remain close at hand. The

archaeological investigation of these image ahu always requires the movement of so many heavy stones that, quite apart from the obvious exhibition value of a restored monument, the easiest way to carry out the investigation is to replace stones in their original positions as the work progresses. Thus considerable restoration generally results from the investigation itself. It was the experience of the writers at Vinapu and Ahu Akivi that the investigation can be handled most easily and efficiently this way with the added advantage of producing as a byproduct a monument valuable for exhibition purposes. (Figueroa and Mulloy, 1960; Mulloy, 1961; Mulloy and Figueroa, 1961). Furthermore there is little doubt that the reassembled monuments would present a much more easily solved problem of permanent conservation than would their scattered debris.

Each ahu presents a restoration and conservation problem with individual characteristics.

Many demonstrate sequences of several periods of rebuilding and, while naturally all of these would be recorded in the investigation, the first restoration problem with most ahu would be to determine of which period in the sequence the restoration would be made. This would differ with different ahu according to the nature of the evidence and with the purpose of producing the greatest variety possible of restored monuments.

The restoration of the ahu exclusive of the re-erection of the statues usually could easily be carried out by the same methods the prehistoric islanders used and without heavy equipment, as the writers have already demonstrated. The work could be greatly accelerated and efficiency would be greatly increased by the use of a light crane attached to the back of a heavy truck of the type used to move wrecked automobiles.

The re-erection of the statues on their pedestals is a considerably more difficult problem though not an insurmountable one. At Ahu Akivi,

seven statues each somewhat over four meters high and weighing about 15,000 kilograms were re-erected by the same methods used by the prehistoric islanders - that is by levering them up and gradually building a mound of stones and earth under them which eventually permitted them to be turned into position. (See Photographs Nos. 3-4). In the future, this method might be used for some smaller statues, but for larger ones the time required would be prohibitive. A heavy crane capable of lifting at least 50,000 kilograms to a height of fifteen meters would be necessary.

Some statues are broken usually into two or three pieces. These too can be restored by binding the parts together with large bronze dowels. This was done successfully at Ahu Akivi to the broken heads of four of the statues and eroded parts of the joints were filled with a cement made with pulverized Rano Raraku tuff which after a few years of weathering satisfied fully the restoration principle that such patches should be visible on inspection but not unsightly. (See Photographs Nos. 5-6).

The largest statue which rested on an ahu and which might be an object of restoration is that at Ahu Te Pito Kura on the north coast to the west of Anakena. This is about ten meters high and is broken into two pieces. It has been estimated to weigh about 82,000 kilograms. (Smith, 1961, p. 203). Few others approach this in weight. The average weight of statues on ahu is probably from 25,000 to 45,000 kilograms. Some are much smaller.

The greatest engineering achievement of the prehistoric islanders was not the transportation and erection of the statues as is frequently stated, but the placing on the heads of some statues of large cylindrical hats or top knots of red scoria carved at the quarry called Puna Pau near Hangaroa. Archaeologists are not sure how they were placed in position. These frequently weigh as much as 5,000 kilograms and the statue at Te Pito Kura has one that probably weighs about 12,000 kilograms. (Smith, 1961, p. 203). Happily, most statues do not have these hats but when they are involved they will present a difficult restoration challenge which

will certainly require the use of the previously mentioned heavy crane. (See Photograph No. 13).

Other kinds of ahu which probably include about 200 examples will rarely prove difficult restoration problems as they are usually of small stone construction. Among these are the semi-pyramidal ahu which are essentially long, narrow mounds of stones highest in the center and sloping toward pointed ends. They have a nearly vertical wall on the sea side and a sloping ramp on the other. Some reach a length of over 100 meters. They contain single tombs or series of these and other burials are found in their mantles. From time to time they are found erected over image ahu.

Others are rectangular structures with flat tops sometimes several meters high and occasionally reaching a length of over forty meters. One or both ends of some examples are elevated. These usually contain tombs and some also have burials in their mantles. (See Photographs Nos. 8, 16).

A numerous and varied class of small rectangular or irregular structures are usually called ahu

avanga. They generally have one or more rectangular cysts in their upper parts. Some contain bones while others appear to have been cremation cysts. Many of the latter kind are associated with image ahu. All of these minor kinds of ahu are of small stone construction and are capable of easy restoration. They do not present special engineering problems. (See Photograph No. 31).

Domestic Structures: The long, narrow, boat shaped, thatched houses with precisely cut basalt foundations and extensive boulder pavements around their lateral doors remain as foundations most frequently on the landward side of ahu plazas though they occur with some frequency in other localities also. They have never been counted but must number several hundred. Their restoration is simple and consumes little time. After investigation, it is generally only necessary to reseat displaced foundation stones and replace the boulders of the outer pavement in their original pattern. (See Photograph No. 22).

Usually at greater distance inland from the

ahu, simpler houses of similar shape and lacking foundation stones are manifest only by their external pavements or other features. These also number in the hundreds and restoration, where indicated, is simple.

Characteristic of many of the domestic establishments are stone chicken houses (hare moa) which also number in the hundreds. These are rectangular small stone structures usually about eight by two meters in extent and two meters high. They contain small longitudinal chambers and have lateral entrances, large enough to admit a chicken. According to tradition, these massive stone structures were built in this manner so that chicken thieves would find their penetration a time consuming and noisy task. They are usually in a bad state of disrepair but their restoration is a rapid and easy task. (See Photograph No. 30).

Frequently associated with domestic establishments and sometimes to be found elsewhere are individual or clustered circular enclosures of masonry called manavai. Most are said to have been

used for the protection of bananas, mahute (paper mulberry), and probably other growing crops. They are called manavai and vary in size from about two meters in diameter to about twenty meters and from one to two meters in height. Frequently they occur in large clusters of a variety of forms with the intervening spaces filled with rubble and appear as massive and impressive structures. Sometimes they are circular masonry lined pits. Restoration consists of the simple replacement of fallen stones and presents no engineering problem. When these are restored, they should be replanted with bananas and mahute which will continue to grow wild and will provide a pleasing appearance for exhibition purposes.

Miscellaneous Monuments: There are more than twenty-seven tower like structures locally called tupa. These are sometimes built over caves and occur in a variety of forms. Most common is a circular flat topped tower about three meters high and five or six meters in diameter with a small side entrance and an interior room. One is built in the form of a

whale and there are other unusual forms. Their former function is unknown. They may have been dwellings or may have served some more obscure purpose. Restoration is usually a simple task of replacement of fallen stones, though in some cases the cantilevering of large stones in roofs requires skill and care.

Also numbering in the hundreds are solid tower or cairn-like structures called pipihoreko. These tend to be cylindrical or slightly conical with flat tops and most vary from one to two meters in diameter and from one to two meters in height. They frequently occur in a linear distribution and traditionally are said to have been boundary markers. Some of them at least contain burials. Restoration consists of the replacement of fallen stones. (See Photograph No. 29).

Extending over parts of the island are the clearly evident remains of prehistoric roads. According to tradition, these were used for the transportation of statues from the quarry at Rano Raraku to the various ahu. This would seem to be true in part at least for near the quarry statues presumably in process of transport are scattered

thickly along the road sides while they occur in similar locations occasionally in other parts of the island. Roads have been noted most clearly by the writers along the south and northwest coasts but less evident remains occur elsewhere. They appear for the most part to have been made by the removal of stones and the smoothing of minor irregularities. In some cases along the northwest coast where the road passes along a steep slope there is some suggestion of dugway construction and of stone walls supporting arroyo fills. In one place inland of Hanga Kihi Kihi, a section of road appears to have been bordered with stone blocks.

The investigation and plotting of these roads will be a difficult problem for long sections of them have been used in modern times to the extent that their prehistoric characteristics have been entirely obscured. The rapid increase of road grading, road use, and new road construction make the investigation of and plotting of the locations of the old roads one of the more urgent parts of the archaeological inventory. Construction of new

roads should be carried out with the idea of leaving as many lengths of the prehistoric roads intact as may be possible. Restoration is essentially the removal of recently scattered stones, occasional repair of fill retaining walls, and removal of minor debris from dugways.

Caves: Rapa Nui has many hundreds of caves of which some are features of spectacular interest. The simplest of these which are found in many parts of the island are small recesses in which twenty to thirty persons might be housed. Many have floors artifically flattened and sometimes terraced to provide living space and entrances narrowed to small doorways by masonry walls. Traditionally, they are said to have been occupied during periods of over-population, but many were undoubtedly occupied at other times also. In many cases, their archaeological investigation will present time-consuming as well as particularly significant problems as their floors typically consist of sometimes deep stratigraphically deposited middens. Their restoration is generally simple and would consist of the rebuilding of floor

terraces, entrances, and interior walls. Caves of this class, after restoration, would be not only monuments of high interest for exhibition, but many would also serve as rain shower or overnight shelters for guided groups of tourists who might be interested in spending a night in a prehistoric Rapa Nui habitation cave.

Another numerous class of caves is traditionally said to have served as refuges from enemies though in many cases it is difficult to understand how they could have been so used. Many probably had obscure purposes not now understood. They are characterized by small passage entrances formed of masonry in great variety which sometimes lead into extensive caves. One example near Hanga Kihi Kihi begins with a rectangular masonry lined shaft about two meters deep which leads into a horizontal masonry lined drift about ten meters long and thence to a second shaft shaped rather like a cork screw which descends another four meters and opens into an extensive cave that has never been thoroughly explored to the writers' knowledge. Another in the central part of the west coast

has a small masonry lined entrance shaft which opens into a natural horizontal volcanic tube which extends about fifty meters and, in turn, opens into two open chambers high in the side of a steep cliff with a breathtaking view of the sea and rocks below. The number and variety of caves of this class is great. Some are volcanic tubes several hundreds of meters long with underground pools and large chambers and a great many other features of interest to speleologists and those interested in cave exploration. Many of these have never been thoroughly explored.

Many have been used for burial purposes and a considerable number of skeletons are still to be seen in them. Though most of this class do not appear to have been suitable for domestic quarters, except in some cases near entrances, it is possible that in certain localities extensive midden deposits may be encountered which will require time-consuming investigation.

Restoration would require a considerable amount of rebuilding of masonry walls around entrances and in interiors and the removal of some interior rock

falls. Most caves of this class could be entered safely by the public only in parties and under the supervision of experienced guides.

Petroglyphs: Scattered widely over the island are thousands of examples of excellent petroglyphs sometimes in large groups and frequently excellently executed in high relief. Some are in caves while most are on large flat, horizontal lava surfaces in the open. They form a significant class of objects for exhibition purposes and require little in the way of conservation measures except minor cleaning. (See Photograph No. 27).

Monuments Which Present Special Restoration and Conservation Problems

Orongo: This unique ceremonial village lies in a truly spectacular site on the narrow crest of the western lip of the crater of Rano Kao. It is bounded on the west by a steep cliff which descends to the sea below and which provides a breathtaking view of the three islets called Motu Nui, Motu Iti, and Motu Kao Kao. On the east, an almost equally steep slope descends to the surface of the kilometer wide

totora covered crater lake of Rano Kao some 200 meters below.

The village, according to tradition, was occupied in prehistoric times for only a short period each year during a part of the month of September by representatives of the various lineage groups which occupied the various parts of the island. A ceremony was held which involved, among other things, a race in which the contestants descended the cliff to the sea and swam to the islet of Motu Nui some 1400 meters away. The first to return to the village with an egg of the manutara (Sooty Tern) was declared winner and his sponsor then occupied during the following year a position of special ceremonial importance.

The village itself occupies a narrow part of the crest some 750 meters long wherein are located about forty-seven houses and an ahu.

These houses are of special construction, not found in other villages on the island. Without considering the many minor variations and special features each is a long narrow oval with walls of

dry masonry made principally of thin slabs laid horizontally. Roofs are domed and made of large slabs cantilevered and carefully interlocked to join at the center. Entrances are very small lateral crawlways. Most houses are built into slopes in such a manner that one side is below ground level. (See Photograph No. 28). Some houses contain paintings and other interior features of interest.

At the southern end of the village is a considerable area of rock outcroppings which are covered with many large and excellently executed petroglyphs done in high relief. There are many figures of the tangata manu (man wearing a bird mask) traditionally said to commemorate the man who won the special ceremonial position by having had his representative win the egg race, figures of the God Make Make, and many others. (See Photograph No. 27).

At the north end of the village is the small ahu associated with a kind of sun dial so constructed as to mark the day of the equinox and the summer and

winter solstices.

This village was partly excavated in 1955 by Ferdon (1961) and will require considerable additional investigation work before it is thoroughly understood. As it is a spectacular archaeological monument, its complete restoration should be carried out as soon as possible. At various times in the past, sheep and other animals have been able to enter the area and have caused many houses to collapse by walking on their roofs. Others have been partly destroyed by natural causes. These can be excavated, investigated and restored by replacing their original materials.

As the entrances of the houses are very low and narrow passages, many visitors would be unable or unwilling to enter through them. It is therefore recommended that some of the collapsed houses be restored with part of the roof left unfinished so that their interiors and roof construction techniques can be observed by visitors standing on the slope above them.

The ahu was thoroughly investigated by Ferdon

and its restoration would require only the replacement of some displaced stones. The petroylyphs require no restoration.

The writers estimate that two archaeologists and about twenty-five local laborers could thoroughly investigate and restore this village in about one year. The site is now isolated from animals by a fence and it is of utmost importance that this be checked frequently and maintained in an excellent state of repair.

The Quarry at Rano Raraku: This is the extensive stone quarry from which most of the island's, probably over 1000, statues were carved. The volcano itself, including its slopes, occupies a roughly circular area some 1,100 meters in diameter and rises to a maximum height of about 100 meters above the surrounding terrain. With its large totora covered lake, it would be a feature of outstanding interest even if the statue quarry were not present. With the quarry it becomes, by all odds, the most magnificent single monument on the island and the classic symbol of Rapa Nui, as known to the world

at large. (See Photographs Nos. 17-18).

The area of archaeological interest involves both the outer and inner slopes of about one quarter of the circumference of the volcano on the south side, being an area about 800 meters long and of varying width. The higher parts of this near the crater lip are formed of exposed tuff and it was from this that the statues were carved. Carving was done by excavating channels around and under each statue while leaving the latter attached to the living rock until the carving, including even fine detail, was almost completed. The statue was then broken loose from a narrow keel along its back and lowered down the slope. This upper area is thus a maze of quarried faces bounding areas from which statues have been removed and has within it about 200 statues illustrating every stage of the carving process including examples ready to be broken from their keels and one already severed and blocked in place with stones. The largest of these unfinished examples is about twenty meters long. Many other unfinished statues will probably be

found when debris is cleared away. (See Photographs Nos. 19-20, 25-26).

Although many statues obviously were lowered and thence transported to ahu, about eighty, mostly very large specimens, remain standing on the lower inner and outer slopes partially buried in talus. Although it is not clear at present whether these latter examples were awaiting transportation or whether they were intended to stand permanently in their present positions, the latter possibility seems likely as many are larger than any ever successfully transported to ahu. Many of these are now buried in talus to their necks and others to near the tops of their heads. Almost surely there are many other examples that are completely buried. On and beyond the outer talus slopes are about thirty prone statues which may have been in the process of transportation. (See Photographs Nos. 21-24).

The problem of successful investigation and preparation of this monument for most effective exhibition is the most difficult one of the whole

island. Skjölsvold carried out a preliminary investigation which cast considerable light on the nature of the problem (1961a), but a great deal more must be learned before any firm plans for preparation for exhibition can be formulated.

Most important is to learn the nature of the talus that so thoroughly buries the standing statues and the conformation of the surface on which they were originally placed. It is probably partly of natural slough from the slope above and Skjölsvold's excavations indicated that much of it is quarry debris. Clear evidence can be obtained only by excavating a series of parallel exploratory trenches transversely across the talus slopes from the flat ground below to the quarry area above. These might have to be as much as four or five meters deep in some areas. How close together they would have to be could be determined only after experimentation.

Once the conformation of the original surface is known, plans might be made for the removal of the hundreds of thousands of cubic meters of earth

necessary to restore the original surface and to present the standing statues in full view, or some other means of dealing with the problem might be worked out.

This problem is so vast and its details remain so doubtful at present, that the writers recommend that it not be carried out as one of the first projects of any long term island wide program of conservation and restoration, but that it await an extended period of mature study and reflection. The monument is an impressive one for exhibition purposes almost as it stands and can be conserved effectively with minor repairs such as the cleaning and removal of minor debris from the statue carving area, the re-erection of certain statues that have fallen recently, and the filling of a number of unsightly holes resulting from former excavations. It is estimated that these minor repairs might be done by an archaeologist working with twenty-five islanders in about fourteen months.

It is noteworthy that, as a result of over grazing, not only the area of the quarry itself, but

all of the exterior and interior slopes of the whole volcano have recently been sadly eroded, and much of the grass cover destroyed. This situation should be regarded as critical and if serious measures are not taken at once, the value of the best known monuments on the island will be seriously impaired and much of the natural beauty of the volcano as a whole will be lost. Essential among these measures is the construction and maintenance of a fence surrounding the whole volcano to insure that no animals enter. Probably the services of an erosion control expert will also be necessary.

The Quarry at Puna Pau: Just northeast of the village of Hangoroa lies the small volcanic cone of Puna Pau. (See map). Here the previously mentioned cylindrical hats or top knots that adorned many statues on ahu were carved from the red scoria which forms the cone and was accessible from within the crater. The cylinders were carved in place, broken loose, and rolled over a low side of the crater and thence to ahu locations. At present, the crater is

about fifty meters in diameter and its interior characteristics are covered by a deep mantle of quarry debris. Several partly finished hats project from this. If the debris were removed, a considerable amount of interesting evidence of quarrying and more unfinished hats would probably be revealed. How difficult a task this would be, with its accompanying archaeological investigation, is impossible to determine without preliminary testing, but with trucks to carry away the debris it might be of the order of six months for an archaeologist and twenty men. It is noteworthy that the pulverized scoria that would be removed is a valuable material for covering roads, pathways and for other purposes. All of it would be used in Hangaroa.

Outside the crater is a prepared inclined road down which the hat cylinders were rolled. This might need a little restoration and investigation but it does not appear to represent an extensive task. Along this roadway are about fifteen large cylinders apparently awaiting transport. These need no

conservation attention. There were formerly many more of these, but many have been rolled to Hangaroa to be cut up for building materials. This practice has now been discontinued, but an important part of the value of this monument has been lost. (For additional information, see Skjölsvold, 1961b).

The Cave called Ana Kai Tangata: This large erosional recess just above sealevel lies on the southwest coast south of Hangaroa and near Mataveri o Tai. It is traditionally said to have been a location where cannibal feasts were held and the name means "cave where men are eaten."

A large part of the roof and walls were once covered with paintings principally of tangata manu (men wearing bird masks). Most of these have been destroyed by time and vandalism but enough remain to form an impressive mural. These should have the services of a specialist in mural restoration to recreate the brilliant colors. It would be worthwhile to bring one to the island for this purpose and for restoration of paintings at Orongo. Time required cannot be estimated but should not be

excessive.

The Poike Trenches: This monument lies just above the foot of the western slope of Poike and extends almost the entire width of the island at this point. (See map). Here are located a series of some twenty to thirty elongated and separated depressions each about 100 meters long. They range from almost imperceptable depressions to others that reach several meters in depth. The dirt resulting from these excavations is piled in a long mound on the uphill side of each.

This feature is the traditional location of a prehistoric battle between two groups and by some observers is thought to have been constructed as a fortification. Others regard it as a feature originally made to collect water from the hill sides to provide moisture for banana plantations in the trenches, and which may have been secondarily used as a fortification on occasion.

Smith (1961b) made test excavations in 1955 which revealed evidence of considerable fire in the trenches. Additional excavations are necessary to

be sure about the nature and purpose of the feature.

The area is suffering badly from erosion and the first conservation measure should be to refill eroded spots and re-establish grass cover. To do this it will probably be necessary to fence the area. If, after investigation, it is determined that the trenches were actually made for banana plantations, the possibility should be considered of removing accumulated debris and replanting with bananas.

Statue Preservation: The Rano Raraku tuff of which most of the statues are made is of very uneven quality. Apparently most of the statues, when new, had smoothly ground surfaces. Because stone from different parts of the quarry differs considerably in its resistance to erosion, because some statues have been in more exposed locations than others, and because some are older than others, there is a great difference in the extent to which they have suffered through time. Smooth surfaces are still present on many buried statues. On all exposed ones the smooth surfaces have been to a greater or less extent eroded away. Beyond this there is great

variation in the extent of erosion with some examples having been so sadly reduced as to be hardly recognizable as statues while others have suffered very little.

The most damaging agent in the erosion process seems to be rainwater which appears to remove the cementing material in the tuff and reduce the surface to an easily detached sandy consistency which condition penetrates progressively deeper.

It will be highly desirable to study this problem and experiment to determine a chemical agent which will halt or inhibit this erosion process without damaging the statues or changing their color. Mulloy obtained a large amorphous piece of the Rano Raraku tuff in 1963 and experimented with several commercial stone preserving chemicals. It would seem that any one of several silicone compounds available on the market are likely possibilities. These inhibit the entry of water without hardening the surface. In cases where surface hardening is desirable the use of polyurethane resins such as those manufactured by

the Texas Refinery Company at Fort Worth, Texas seems a possibility. Much more experimentation is necessary before any chemical is used. As the wrong material might be very destructive, restricted initial use and extended observation of the effects of time before any widespread application is indicated.

The Principles of Restoration

The character of the Rapa Nui monuments is such that the writers recommend that restoration be principally of the type known as anastylosis or reassembly of the original materials that remain at hand. The introduction of new materials should be kept to a minimum and these should be used only when necessary for purposes of stabilization. From time to time this will be necessary as at Ahu Akivi where it was considered necessary to cast large blocks of concrete within the rubble interior of the central platform to provide stronger support for the reerected statues than the islanders originally used. This was done in such a manner that no surface manifestation of the concrete appeared. Similarly some dry masonry walls were stabilized with concealed concrete. In some

cases it will be necessary to add new stones or other materials where foundation or other supporting elements are lacking, but this should be kept to a minimum. Essentially, new materials should be used only where necessary for the security of superimposed elements.

This will mean that many restored monuments will appear to some extent as ruins rather than as they appeared when originally constructed.

Salvage Archaeology

There are several localities where proposed construction activities or other immediate dangers make conservation and restoration work or investigation of monuments in advance of their destruction by essential construction urgently necessary.

One of the most urgent of these is in the town of Hangaroa and its environs. In this area between Mataveri and Tahai (see map) there are at least ten large and important ahu and about six smaller ones as well as a number of other kinds of structures. Quite apart from their scientific and historical

value, these are potentially fine monuments and significant future assets to the town. As there are plans for much construction here in the near future and as these are in a somewhat flexible state of development with many changes probable before the construction actually takes place, the most effective means of solving the problem would be as follows: All of the monuments in this area should be investigated and restored as soon as possible. Once this is done, the authorities in charge of construction and planning will become aware of them as the town assets they are and will plan other construction in relation to them. They will thus become an harmonious part of the growth pattern of the town.

An alternative would be to complete the archaeological inventory in this area as soon as possible and reserve the land on which monuments are located. This is less desirable because the harmonizing of new construction with an existing monument is easier and more likely to be carried out than with a potential one.

Another area of urgency is that between Anakena and Ovahe bays on the north coast. (See map). (See Photographs Nos. 11-12). According to Peterson's assessment of island resources this area is by far the most likely to be used for the construction of a resort hotel and related tourist facilities. In such an area to be devoted to tourists, the harmonious relationship of archaeological monuments to new construction has especial value and there are many monuments here. In this general area, there are at least sixteen ahu as well as houses, caves, tupa, chicken houses, manavai and other monuments. The large hill called Maunga Hau Epa has an artifically flattened top and terraced sides and is a prominent monument. Here again the immediate investigation and restoration of all the monuments in the area would, in addition to the historical information that would result, not only serve to conserve them but with new construction adapted to their locations the resort value of the area would be increased greatly. Such new construction adaptation could be carried out only after

the restoration of the monuments.

Within this zone, one sector is of especial urgency. Peterson selected the area to the north of Maunga Hau Epa and Maunga Te Puha Roa as the most advantageous site for a resort hotel building and it is probably the only truly suitable one in the area. It lies by the sea with easy access to the beaches at Anakena and Ovahe and buildings here would not produce an incongruous contrast with any of the large restored ahu. This extensive and picturesque area has a great many small monuments including two tupa, two small and inconspicuous ahu, several caves and a number of chicken houses and manavai. They are so situated that a unique and picturesque resort hotel could be designed to take advantage of the terrain among them while leaving the monuments intact as decorative features. The many manavai could be replanted with bananas and mahute and would provide especially decorative and characteristic background. In order that a designer could take advantage of these unique features, it would be necessary that they be first restored to

their original form so that new elements could be harmonized with them. It is thus recommended that archaeological investigation and restoration of the monuments in this sector be carried out in advance of other parts of the Anakena-Ovahe area and before the hotel is designed.

Some sort of salvage operations should also be carried out in advance of construction of new air-strip facilities. Two areas have been surveyed. One extends between Hotu Iti and Anakena. The other follows the line of the existing airstip facilities from Mataveri to Vinapu. (See Peterson's report for more details of proposed airstrip construction). It is not clear as yet which of these sites will be used and archaeological salvage operations will depend on the eventual decision. Time did not suffice for the writers to make detailed surveys of the proposed sites of these two airstrips and this should be done as soon as it is firmly decided which will be used. It appears that there are at least two ahu of which one at least is an important monument with unusual statues (Unnumbered ahu

between Ahu Nos. 87-88) as well as several chicken houses and other small monuments within the area of the Anakena-Hotu Iti strip. If this strip is eventually used, attention should be given to the idea of moving at least the important ahu and perhaps other monuments as well to a safe area outside the strip. As the statues and platform of this ahu are small, it could be moved and its investigation carried out by an archaeologist and twenty men with suitable heavy equipment in about six months. Other monuments in the area would provide much smaller tasks.

The writers know of no important monuments within the area of the proposed Mataveri-Vinapu strip, though a detailed archaeological survey should be made to assure this. It is possible that operations will cut into the side of Maunga Orito, the obsidian hill from which prehistoric islanders obtained much material for stone tools. This might uncover significant evidence that would require a salvage operation.

The area of Hotu Iti and its environs will

surely be the site of extensive port construction if the Anakena-Hotu Iti airstrip is built. Even without the airstrip construction, this area will probably be the site of some construction in the near future. It thus represents a salvage operation of some urgency. There were originally about nineteen ahu in this zone but in 1960, a most unusual and destructive tidal wave passed over this area carrying in this limited sector a wall of water perhaps seven or eight meters high as much as two hundred meters inland. Several of the ahu and certain other monuments were destroyed beyond any possibility of restoration. The remaining ahu that are still capable of investigation and restoration include Nos. 150, 151 and 154. Though these lie to the east of most probable construction, they should be investigated and restored and new construction adapted to them. No. 154 in particular has two statues and might be especially significant. These four ahu could be investiaged and restored by an archaeologist and twenty men in about six months with the help of a crane.

It is noteworthy that Ahu No. 158, Ahu Tongariki, was among those destroyed by the tidal wave. This was the largest on the island, having been about 150 meters long with a high platform wall made of precisely cut large stones. It had fifteen statues that probably weigh about 50,000 kilograms each. The ahu was completely destroyed and its walls and statues were thrown inland, some statues to a distance of over a hundred meters. Though it conceivably might be possible to restore this ahu to something like its original form by making use of the many detailed photographs that exist, the task would be a tremendous one and with the large number of other fine ahu that could be restored at a fraction of the cost, the value of a restoration project seems questionable. It is recommended here that the debris of this ahu be left as it is at least for the present and that the area be reserved for exhibition as an illustration of the power of a tidal wave. The area to landward of the original site of the ahu is now an assemblage of 50,000 kilogram statues, large and beautiful cut stones, and rubble. It is spectacular

and of much exhibition interest just as it stands.

Physical Anthropology

Ten years ago there were hundreds of skeletons to be seen in caves, ahu tombs, and in other locations in many parts of the island. Since that time the writers estimate that at least three-fifths of these have disappeared. Many have been carried away by tourists, some have been eaten by animals, and others have been carried to the continent and sold by the local people. Such material of this kind as is visible on the surface is rapidly being lost but much remains. When archaeological investigations begin, if previous experience is any index, a great many now buried skeletons will appear. As cremation appears to have been common during a considerable period of the island's history, most of these will probably involve late prehistoric and historic horizons, but it is not certain that earlier material will not also appear.

There thus exists the opportunity to collect and maintain in a single repository an unusually large and most valuable series of skeletons which

would be added to continuously as archaeological work progresses. In order to assure the preservation of this valuable evidence it is essential that a physical anthropologist be present on the island at least as soon as the archaeological work begins and preferably sooner. The first task of this individual would be to collect all of the visible skeletal material, record its provenience, apply chemical preservatives where necessary, catalogue, and store it in such a manner that it would be accessible to students. As excavation progresses he would have the further task of supervising the excavation and preservation of additional skeletal material and similarly processing it.

Such a physical anthropologist could also devote a part of his time to the study and measurement of the living population. Valuable contributions to knowledge would result.

Social Anthropology

It is noteworthy that Rapa Nui at this moment provides a unique opportunity to carry out one phase of a series of social anthropological studies

of considerable potential value. Alfred Métraux in 1934-35 produced an excellent ethnographic description of native Rapa Nui culture of the contact period principally by searching and evaluating the memories of informants of his time. (Métraux, 1936). Since the contact period, the people of the island have been subjected to outside influences of gradually increasing intensity. These have been brought by the relatively rare arrivals of ships from many outside areas, from the Church, from the influences of English and Chilean sheep company employees who remained for considerable periods in residence, from Chilean schools and Chilean officials in more recent years, and from periods of residence in the Continent of probably about a third of the population during the last ten years. The old culture has thus been highly modified by relatively slow diffusion type contacts which have had a reasonably good opportunity to become integrated into the operating pattern. If proposals for the development of tourism and intense ship and air contact in the near future develop as planned, the nature and intensity of outside contacts

will change sharply and become characterized by all of the typical personal and social traumata of an acculturation situation. The present thus represents an essential turning point in Rapa Nui culture and it would be of great value to have a detailed description and evaluation of it as it now exists. This, together with a similar investigation perhaps ten years hence would be of general theoretical value in the study of culture change and would also be of considerable practical use in dealing with and attempting to prevent or ameliorate the personal traumata that are sure to arise with many of the local people under the impact of acculturation forces. Such an investigation could be made by a social anthropologist without assistance in the course of a year's residence.

Museum

A most necessary aspect of the foregoing proposals is the construction of adequate museum buildings for the storage, study and exhibition of the large number of cultural artifacts and skeletal material that will result from the

investigation program. Such a museum could be constructed in such a manner that it could be enlarged as needed. It should include adequate exhibition space, a large amount of storage space, a photographic dark room, laboratories for study, preparation, and cataloguing of materials, and an adequate library to permit comparative studies. As time passes efforts should be made to obtain samples of comparative materials from other parts of Polynesia so that much of the work of preparation of the large number of reports that would result from the investigations could be done on the island. This museum could eventually become one of the important centers for the study of Polynesian prehistory.

The Chilean Ministry of Public Works is in the process of designing a museum exhibition hall to be constructed at Apina in Hangaroa. (See Map). For the purposes here proposed, additional buildings will have to be added. An area large enough for future needs should be reserved in the vicinity of Apina.

The Organization of an Investigation, Conservation and Restoration Project

As the organization of such a project must

necessarily involve a great many factors not known to the writers at present and will require considerable evolution through time, a consideration of certain initial requirements only may be approached here in the most general and minimal terms. It is clear that the task is so vast that such a project, once initiated, should continue into the indefinite future and its form and functions would be expected to change as different aspects of the problem are approached.

The project must be administered by a recognized scientific institution operating in the field of Anthropology in order to establish broadest policy, provide an outlet for the many publications that will result, and to provide general access to the scientific world. Such an institution as The National Museum of Natural History of Chile would be ideal.

In the past the most difficult problem involved with archaeological work on the island has been isolation from continental contacts and sources of supply. In 1960, for example, the writers were isolated on the island for a full year. It is understood that the Chilean government plans to

send about one ship every two months to the island during the coming year and that the airstrip will be completed and scheduled flights begun as soon thereafter as possible. The following proposals are made with this assumption.

In spite of the urgent necessity for salvage and inventory work, it would probably be most reasonable to begin the project as modestly as possible and, as success is demonstrated and results are desired more rapidly, to add to it as is indicated.

The first essential is a director who should be an archaeologist familiar with Pacific problems, field work and restoration techniques, and also with Chilean culture and the Chilean governmental agencies with which he would cooperate. He would be charged with making plans and policies, obtaining of personnel and supplies, supervision of the work, insuring preparation and publication of results, and the general development of the project. He would probably be able to supervise directly some of the work in the early stages of development but it would be expected that eventually his work would become entirely

administrative. He should be aided by a business manager and a secretary.

As the archaeological inventory and salvage work are clearly the two tasks of greatest initial importance, first attention would be directed to these. Three archaeologists should be obtained of whom one would devote himself to the inventory and two to salvage work - probably first at Hangaroa. A permanent surveyor-draftsman and a photographer would also be necessary. Two additional surveyors might also be obtained from the Chilean Ministry of Public Works on temporary duty to aid with the inventory.

Laborers and labor foremen are available among the local people. About five local men would be necessary for the inventory. For archaeological investigation and restoration, it has been found in the past that a ratio of about ten laborers and a foreman to be supervised by each archaeologist is most desirable. Under some conditions this ratio might change. Thus about twenty-five laborers and two foremen would be indicated initially. As it is clearly most desirable to carry on the salvage work

more rapidly than could be accomplished with these crews, consideration might be given to adding additional archaeologists and laborers at the indicated ratio. The previously mentioned highly desirable physical anthropologist, and social anthropologist should be added if at all possible.

The project should be organized in such a manner that it can make use of anthropologists and other specialists from various parts of the world who are supplied with their own grant or other financing and who may be interested in special projects. Through the years a considerable proportion of the work might be accomplished by doctoral candidates and other fully trained people interested in special objectives. Such work would be expected to be carried on under the supervision of the director and in accord with established policies related to standards of recording, the requirement that restoration or stabilization be carried on as part of the investigatior, and such other strictures as the director might determine. The project should be organized so as to be able

to supply maximum facilities and other aid to such special investigators.

It is also likely that many graduate students from various institutions of the world might wish to engage in supervised archaeological work to fulfill academic requirements. As such people would be expected to come at their own expense or with outside financing, a considerable amount of otherwise unavailable supervisory and other special work might be obtained from them. The project should thus make contact with many universities and attempt to secure such people and be so oriented as to serve to some extent a field school function.

The major items of heavy equipment necessary should include initially two 3/4 ton pick-up trucks for general transportation of equipment and personnel, one two ton truck fitted with a crane and capable of lifting about 3,000 kilograms for moving large stones in ahu restoration (similar to trucks used to move wrecked automobiles), and a dump truck of about four ton capacity to move excavated material. Two gasoline operated cement mixers would be necessary.

Also desirable but not necessary in the initial stages of development would be a small back hoe.

The most expensive and difficult item of heavy equipment is a crane capable of lifting about 50,000 kilograms to a height of fifteen meters at about a three or four meter radius. This would be most useful if rubber tire mounted, though a caterpillar mounted crane would be usable. The vehicle should be mobile enough to be moved to all parts of the island. It is the understanding of the writers that a heavy crane will be brought to the island to be used in connection with the port facilities and, as this item is expensive and requires specialized operation and maintenance, it might be desirable to arrange for its cooperative use with the port authorities. It would, of course, be necessary that the crane selected be mobile enough for archaeological purposes. A good part of small ahu restoration can be done perfectly well without the use of the crane, but for the erection of the large statues and especially the hats, the use of primitive methods will be prohibitively time and labor

consuming.

In case it is impossible to obtain the use of a crane during the initial stages of the restoration work, it would be desirable to have a large light metal tripod about fourteen meters high and capable of lifting about 20,000 kilograms. This should be so designed as to be capable of being erected safely and reasonably easily. In the work at Ahu Akivi, the writers used successfully a somewhat smaller improvised tripod made of eucalyptus poles. (See Photographs Nos. 5-6). The College of Engineering at the University of Wyoming is now working on the design of such a tripod and will undoubtedly produce a usable device.

As the above described equipment will require maintenance and repair, it would seem desirable to develop an arrangement with other agencies to be working on road, port, airstrip and other construction for cooperative repair services. It may even be possible to develop some sort of cooperative motor pool use of vehicles and equipment so that purchase of special vehicles for archaeological use will not

be necessary.

Beyond the items discussed above, the usual smaller tools, equipment, and supplies needed in restoration and investigation work will be necessary but these need not be enumerated here.

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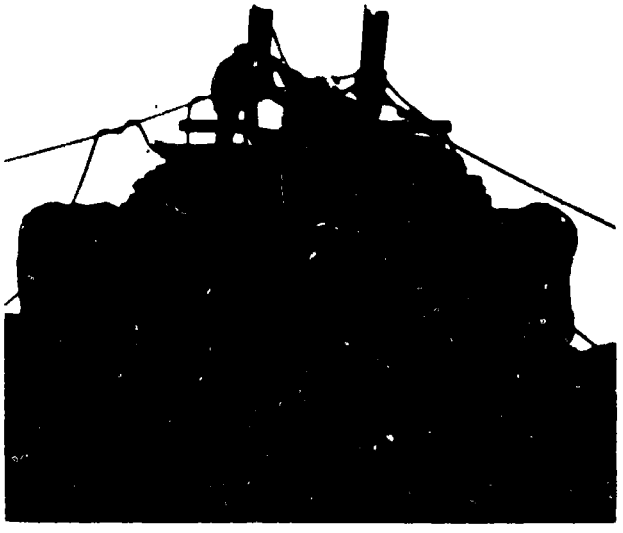
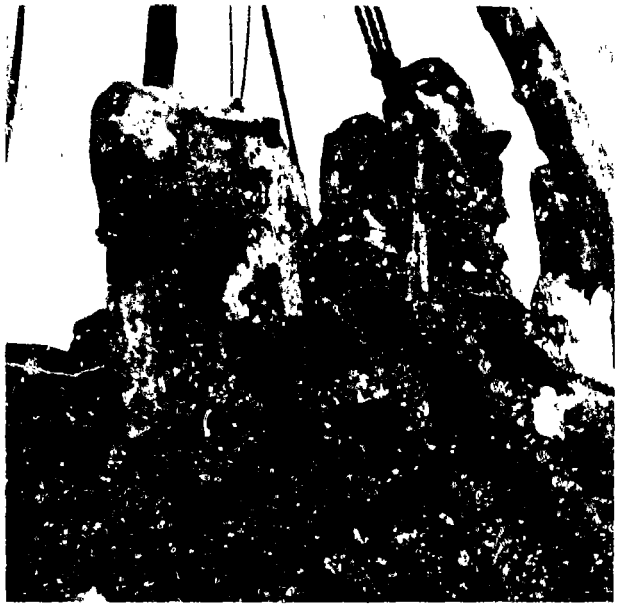
Here *Abu Akivi* is shown as it appeared before investigation and restoration were begun in 1960 by the writers. Only the backs of the seven statues which had been tipped forward in prehistoric warfare and a slight elevation masking the *abu* platform were visible. The monument was so inconspicuous that it was known only to a few people on the island and its true extent was thoroughly masked.

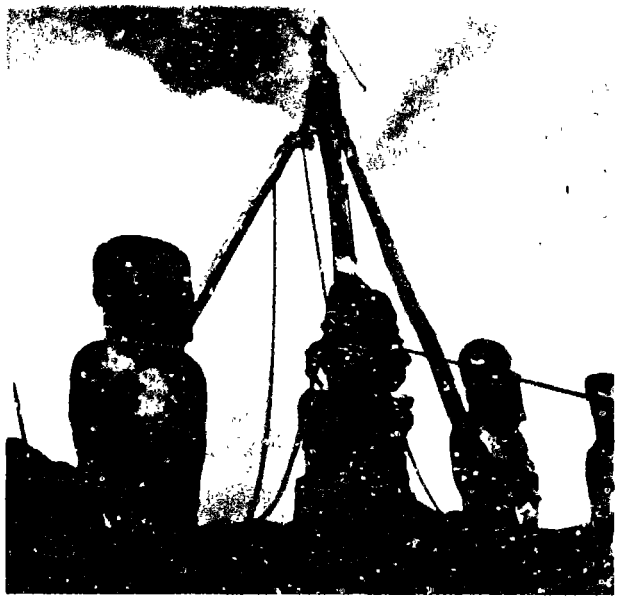
Here *Abu Akivi* is shown after excavation and restoration. In this view is visible only the central platform surmounted by seven statues, each over four meters high. Not visible are the lateral wings that extend the *abu* to a maximum length of about ninety meters. The *abu* faces on a flat plaza about 100 meters square. The monument can now be seen at a distance of several kilometers. Among *Rapa Nui abu* this is a relatively small one. Some have many more statues. Statues associated with *abu* reach a maximum height of ten meters.



The statues at *Abu Akivi* were re-erected by means similar to those used by the prehistoric islanders. The writers first assembled under each statue a frame of heavy timbers projecting beyond the statue to provide seats against which to use long eucalyptus levers. Statues were gradually levered into a position facing their restored bases and then a mound of stones, timbers and earth was gradually constructed under them by levering up the statues by slow stages. Once the statues were nearly vertical opposing cables were attached to the frames and they were pulled into position by manpower. The first of the seven statues required about a month to raise. By the time the last was raised enough had been learned about the technique so that only a week was required. Many *abu* with smaller statues could be restored this way in the future without prohibitive cost, but the restoration of the larger *abu* will surely require the use of a heavy crane.



Shown here is part of the wall of the central platform of *Abu* No. 1 at *Vinapu*. The precisely cut and fitted lava blocks are similar to much Inca masonry. This kind of masonry on the island is one of the elements that have suggested continental contacts to some observers. The question of possible continental contacts can be resolved only after much archaeological research in the future. This *abu* has the magnificent central platform shown, large lateral wings and faces on a large plaza. Six large statues lie prone on its ramp. It is capable of restoration and would be an impressive monument.



At *Abu Akivi*, four of the statues were broken transversely at the neck. The bodies of these were raised as described on the previous page. To raise the heads a small timber frame was attached to each and it was levered into a position upside down in front of its body. In this position two large bronze dowels were seated in the broken surface positioned to fit holes in the broken surface of the body. The head was then tipped over to rest bottom side down in soft earth in a position close to the front of the body. (Photo. No. 5) A tripod of eucalyptus poles was erected beyond the end of the *abu* platform and then levered leg by leg until it rested over the statue body. The head was then raised with a chain hoist while being held laterally with a cable snubbed on a post. At the proper height it was allowed to swing over the top of the body and was then lowered into position. While being supported by the hoist, the eroded juncture was filled with a cement made with pulverized *Rano Raraku* tuff. After five years of weathering, the joint was visible on close inspection but not unsightly. The tripod was then levered leg by leg to a position over the next headless statue and the process repeated. In the future, other small heads could be raised by use of a similar tripod. Much better would be a larger metal tripod of known strength and capable of being erected much more easily and of a type now being designed at the University of Wyoming. For broken heads and hats of larger statues, a crane will surely be necessary.

This is a small well preserved *abu poepoe* of a type frequently found. It contains at least one and probably several tombs. This one lies on the north coast near the foot of *Poike*. It would require investigation but no restoration. Many others of its kind are in a much poorer state of preservation and would require restoration. This could be accomplished easily because of the small stone, dry masonry construction.

This view shows a part of the sea facing wall of *Abu Hanga Pokura* near the western end of the south coast. The masonry here is typical of many large *abu*. Though the walls are partly destroyed, the stones used in them are mostly near at hand and can be replaced easily. The *abu* bears an as yet undetermined number of large statues and could be restored to form an impressive monument.

This view shows a part of the beach area at *Anakena* which will probably be adjacent to a resort hotel. In this vicinity there are seven *abu* of which five are visible here. The results of prehistoric warfare and other agencies have reduced them to amorphous mounds of stone of no exhibition value and little scientific interest. They can be restored at modest cost and, quite apart from the archaeological evidence that will accrue, the area will be converted into one of spectacular interest.

This view shows the setting of *Abu Vaihu* near the center of the south coast. There are eight large statues that lie prone on the *abu* ramp and two hats may be seen lying on the plaza. To the right of the statues may be seen part of a typical small stone mantle placed over the *abu* wing in late prehistoric times to cover burials and perhaps also the beginning of an attempt to convert the *abu* with its fallen statues into a semi-pyramidal *abu*. As the statues here probably weigh thirty to forty tons and they have red scoria hats brought from the crater of *Puna Pau*, a crane would be required for restoration purposes.

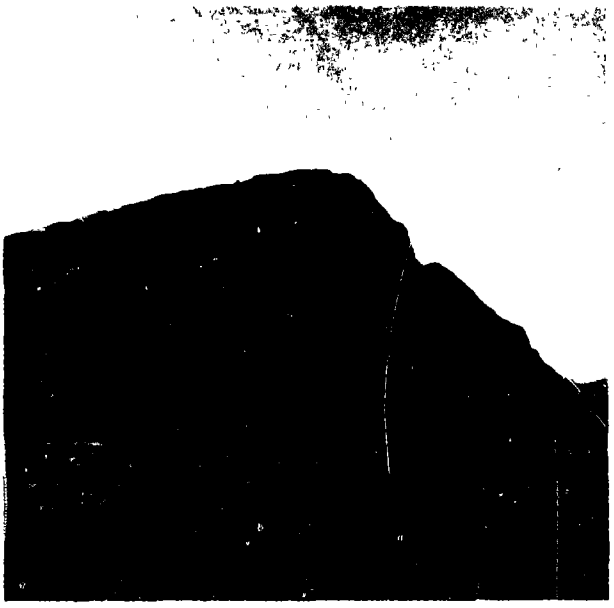
This view shows the present condition of *Abu Nau Nau* at *Anakena*. It has been destroyed by prehistoric warfare and nearly covered with sand. The wall of the *abu* is constructed of precisely cut large lava stones of which some are adorned with carved figures in relief. The *abu* has an undetermined number of large statues and could be restored easily.

This is a view across the plaza of *Abu Vaihu*. Visible are eight large statues that probably weigh thirty to forty tons each. In the foreground are several large hats of *Puna Pau* scoria. Between the statues parts of the plaza side, central platform wall o precisely cut lava slabs may be seen. The stones lying on the ramp represent the beginning of a late prehistoric mantle to conceal burials. This *abu* would require a heavy crane to re-erect its statues.

This view shows part of the seaside wall of *Abu Mahatua* on the north coast near the foot of *Poike*. The wall in this case is made, not of cut stone, but of large selected blocks of which some are more than two meters in diameter. Though it is partly destroyed the stones lie at hand and the wall could be completely restored with the use of a crane. As the ramp of this *abu* is heavily covered by a late prehistoric mantle of small stones, the number of its statues is unknown.

This small, unnamed *abu* lies near the west end of the south coast. It is similar to many others on the island. Its three statues weigh about ten tons each. Though the *abu* has lateral wings, these are so thoroughly buried as not to be visible. As the statues are small and there are no hats, this *abu* could be restored easily without heavy equipment and with methods similar to those the islanders used in its original construction.

This is a rectangular *abu* that is situated on the east central part of the north coast. This one has rounded and elevated ends of which one is destroyed. It may have been constructed to represent a boat. It contains several tombs. There are many others similar to this one on the island. As the construction is of small stones, investigation and restoration are not difficult problems.



This is a view of part of the exterior section of the statue quarry at *Rano Raraku*. Many of the rock surfaces visible on the upper slopes demonstrate cut surfaces from which statues have been removed. Many other partly finished statues remain of which several are visible in this view. On the talus slope in the foreground may be seen a number of partly buried statues in standing positions. Much of this talus which partly buries some statues and almost completely buries others is quarry debris. Selection of a method of preparation for exhibition for this monument depends on a determination of the nature of the talus surface underlying the quarry debris. Eventually it may be desirable to remove a vast amount of this so that the statues may be seen. Probably many statues are completely buried in this talus. It is recommended that this monument not be among the first restored. A long period of preliminary study and investigation should precede any extended restoration activity.

This is a view of the largest unfinished statue in the exterior part of the quarry at *Rano Raraku*. It is about twenty meters long and is more than half finished. There are many smaller examples in the quarry in about this stage of manufacture.

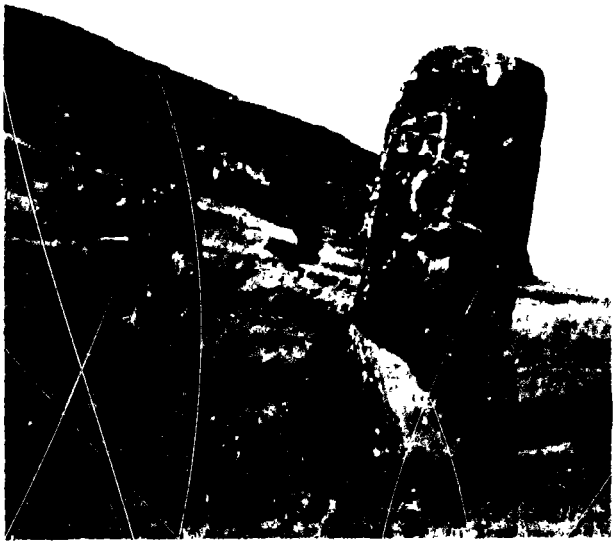
This is a view of the interior of the volcano at *Rano Raraku* showing several statues partly buried in quarry debris and in the background the crater lake partly covered with a mat of *totora*. The crater as well as other parts of the volcano have recently suffered seriously from erosion caused by over-grazing. Much soil has been carried into the lake and more will follow. The erosion problem here is critical and needs immediate attention

The cliff in the lower part of this view is a man made quarry face. In it is a partly finished statue carved from the side rather than from above. This method is unusual but not unique as there are several other similar examples in the quarry. In the slope above may be seen a number of quarried surfaces and unfinished statues.

This view shows a part of the exterior talus with partly buried statues at *Rano Raraku*. These statues are well over half buried in talus that is probably principally quarry debris. They would be much more spectacular monuments if they were completely exposed, but this would require the removal of vast amounts of earth and debris.

This is a view down a part of the exterior talus slope at *Rano Raraku*. Several partly buried statues may be seen with the erosion caused by unfilled recent excavations around them. There is a good deal of this kind of erosion that should be filled as soon as possible.

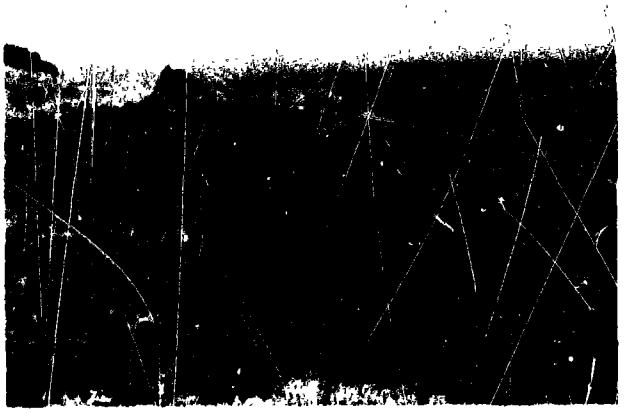
In the foreground may be seen the precisely cut basalt foundation stones and lateral boulder pavement of one of the boat shaped houses that formerly had a thatched superstructure. There are hundreds of these on the island. Many are much larger than this example. Most are clustered along the land sides of *abu* plazas. The quarry situation of this one is most unusual.



These are views of parts of the exterior talus at *Rano Raraku*. The extent to which statues are buried may be seen here.

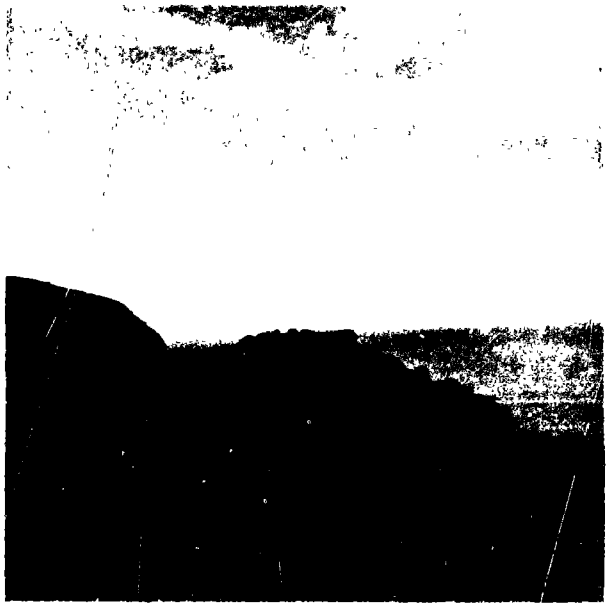
These are views of part of the quarry section on the upper exterior slopes of *Rano Raraku* showing partly carved statues. The upper view shows the method of carving the entire statue while leaving it attached to the living rock by a narrow keel. The lower view shows a finished statue detached from its keel and ready to be lowered down the slope.

This is a view from the south end of the village of *Orongo* showing in the background the islands of *Motu Nui*, *Motu Iti*, and *Motu Kaokao*. It was to *Motu Nui*, the most distant island, that the contestants swam to bring back the egg of the *manutara* (Sooty Tern) in the race that was part of the annual *Orongo* ceremonies. In the foreground may be seen a small part of the large collection of petroglyphs carved in lava rocks in this section of the village. The complete and partially visible figures are those of the *tangatamanu* or man wearing a bird mask. It is said to represent the sponsor of the winning contestant in the egg race. The *tangatamanu* occupied a position of special ceremonial importance during the year after he achieved the position.



This view shows the roof and a part of one wall of one of the elliptical stone houses at *Orongo*. These partly buried structures have lateral entrances. Many of them, unlike this one, are in a bad state of repair and require restoration.

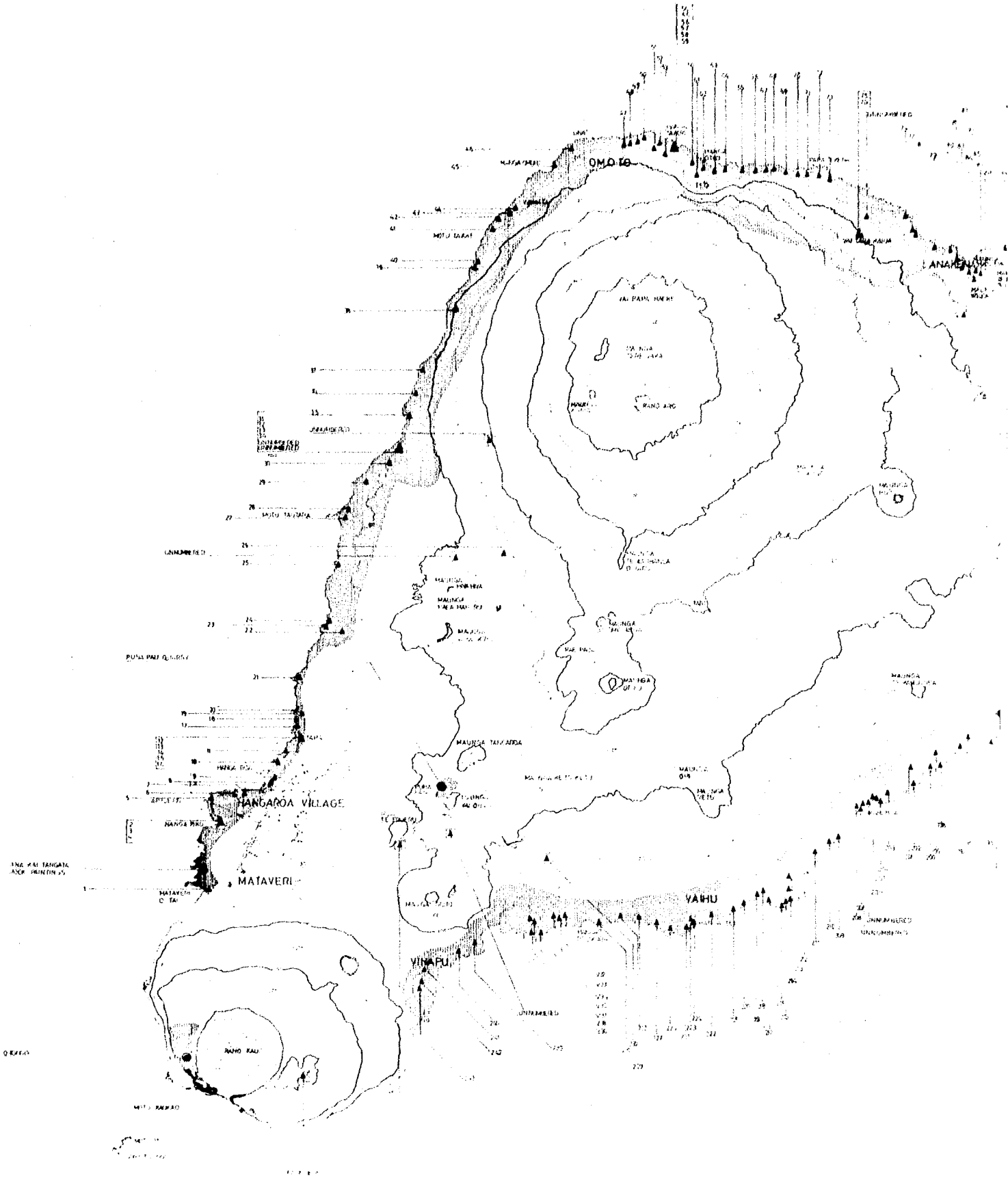
This is a typical *pipiboreko* or cairn near *Abu Vaiteka* near the west coast and inland of *Abu Tepeu*. These are traditionally said to be boundary markers and some, at least contain skeletons. They frequently occur in long lines and their mapping may give some clue to the exact sections into which the island was traditionally said to have been divided. Many are in a bad state of repair, but their restoration is a simple process of replacing the stones lying close by.

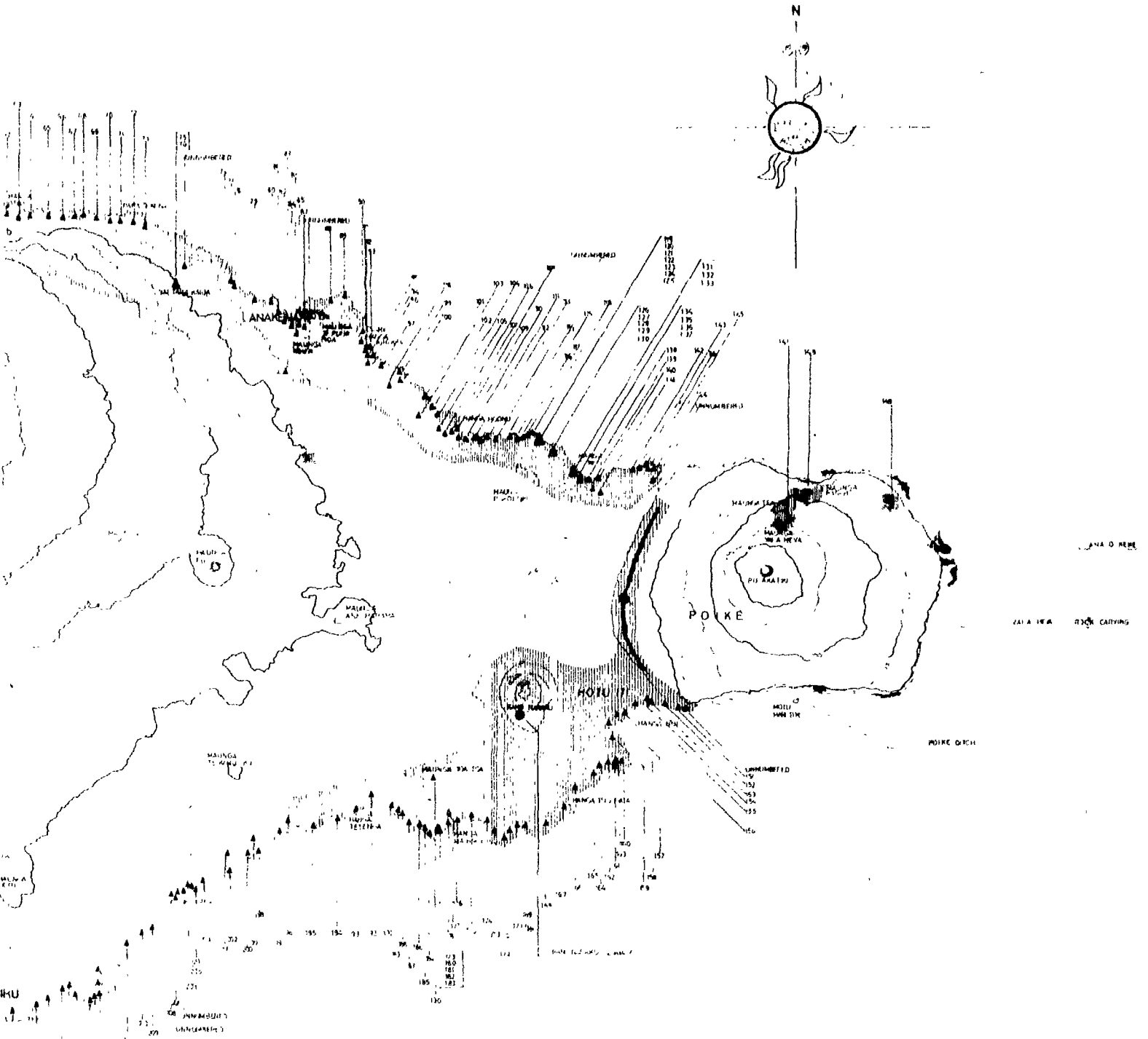


This is an *avanga* near *Abu Matakí te Moa* on the northern part of the west coast. These structures are rectangular or sometimes irregular platforms which bear one or more cysts on their tops. Some of these appear to have been made to contain human inhumations while others contain burned human bones and seem to havé been used for cremations. Many, like this one, are in bad repair but their restoration is a simple process. There are many on the island.

This is a *hare moa* (chicken house). These massive structures contain a small elongated central chamber and a lateral entrance large enough to admit a chicken. There are hundreds of these on the island and they are a characteristic feature of many domestic establishments. Traditionally they are said to have been built in this way so that chicken thieves would find gaining entrance to be a time consuming process and one likely to awaken the owners. Many are in bad repair but their restoration is a simple process

This is the mask called *Vai a Heva* which lies on the side of the small hill *Maunga Vai a Heva* on the upper slopes of *Poike*. It apparently was originally a natural catch basin for water which was incorporated as the mouth in a large relief carved mask said to represent the God *Make Make*. The mouth usually contains water.


OMOTO
ANAMA
TANGAROA VILLAGE
MATAVERI
VAHU
VINAPU
RANO KAU
UNNUMBERED



RAPA-NUI

ARCHEOLOGICAL AREAS

- AHU
- AHU GROUPS
- OTHER SITES